# BASIC COURSE IN MATHEMATICS (MMT022I)


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## SEMESTER - 1st to 3rd

## MULTIDISCIPLINARY COURSE

### MMT022I: MATHEMATICS / APPLIED MATHEMATICS (BASIC COURSE IN MATHEMATICS)

**CREDITS: 03 (HOURS: 45)**

**Note: The external paper will be for first 3 units.**

## Course Overview and Objectives:

1) To aware the students about set theory, real and complex numbers.
2) To understand the basic concepts of coordinate geometry
3) To prepare the students for applying basic mathematics for computational purposes.

### UNIT-I

Introduction to set theory: Sets, Types of sets, Subsets, Basic operations on sets, Power set, Finite set, Infinite set, Countable and Uncountable sets and their examples, Cartesian product, Basic operations, D'-Morgans laws, Relations, Equivalence relations, Partially ordered sets.

### UNIT-II

Real number system, Rational and Irrational numbers, Closure property of reals, Complex numbers, equality of complex numbers, operations on complex numbers, modulus and amplitude of a complex number, polar form of a complex number.

### UNIT-III

Rectangular coordinate system, distance and section formulae, equation of straight lines, various forms, angle between lines. Second degree homogenous equations representing straight lines and angle between them. Matrices and their types, algebra of matrices, determinant of a square matrix.

## Recommended Books

1. Set theory Schaum's series
2. Matrices by Aziz, Nisar and Zargar
3. Complex trigonometry by Aziz, Nisar and Zargar.
4. Matrices by Shanti Narayan
5. Coordinate geometry by Shanti Narayan
6. Mathematical Analysis by S.C. Malik

# Chapter 1

# Set Theory

## 1.1 Introduction to Set Theory

Set theory provides the fundamental language and foundation for modern mathematics. This chapter introduces the basic concepts of sets, their properties, and operations. We will explore how mathematical objects can be collected into sets and how these sets relate to one another through membership, inclusion, and other fundamental relationships. These concepts form the essential building blocks for further study in mathematics and logic.

### 1.1.1 Sets

A set is a well-defined collection of distinct objects. The objects in a set are called its elements or members. Sets are usually denoted by capital letters (e.g., $A, B, C$), and their elements are denoted by lowercase letters (e.g., $a, b, c$).

**Definition 1.1.1** (Set)**.** A set is a collection of distinct objects, considered as an object in its own right. The objects in the collection are called elements or members of the set.

#### Representation of Sets

There are two primary ways to represent sets:

1. **Roster or Tabular Form:** In this method, all the elements of a set are listed, separated by commas, and enclosed within curly braces $\{\}$.

   **Example 1.1.2.** The set of vowels in the English alphabet can be written as $V = \{a, e, i, o, u\}$. The set of even numbers less than 10 can be written as $E = \{2, 4, 6, 8\}$.

2. **Set-Builder Form (or Rule Method):** In this method, a common property that all the elements of the set satisfy is described. It is generally written as $S = \{x \mid P(x)\}$, which reads as "the set of all $x$ such that $x$ has property $P$".

   **Example 1.1.3.** The set of all natural numbers can be written as $N = \{x \mid x \text{ is a natural number}\}$. The set of vowels in the English alphabet can also be written as $V = \{x \mid x \text{ is a vowel in the English}$

#### Membership

If an element $a$ belongs to a set $A$, we write $a \in A$. If an element $a$ does not belong to a set $A$, we write $a \notin A$.

**Example 1.1.4.** Let $A = \{1, 2, 3, 4, 5\}$. Then $3 \in A$ (3 is an element of set A). However, $6 \notin A$ (6 is not an element of set A).

### 1.1.2 Types of Sets

**Definition 1.1.5** (Empty Set)**.** A set containing no elements is called an empty set or a null set. It is denoted by $\emptyset$ or $\{\}$.

**Example 1.1.6.** The set of all real numbers whose square is $-1$: $A = \{x \in \mathbb{R} \mid x^2 = -1\} = \emptyset$. The set of current presidents of the USA born before 1900: $B = \emptyset$.

**Definition 1.1.7** (Singleton Set)**.** A set containing exactly one element is called a singleton set.

**Example 1.1.8.** The set $\{5\}$ is a singleton set. The set of even prime numbers: $P = \{x \mid x \text{ is an even prime number}\} = \{2\}$.

**Definition 1.1.9** (Finite Set)**.** A set is said to be finite if it is either empty or consists of a definite number of elements.

**Example 1.1.10.** The set of days in a week: $W = \{\text{Monday, Tuesday}, \ldots, \text{Sunday}\}$. This is a finite set with 7 elements. The set of solutions to $x^2 - 4 = 0$: $S = \{-2, 2\}$. This is a finite set with 2 elements.

**Definition 1.1.11** (Infinite Set)**.** A set which is not finite is called an infinite set.

**Example 1.1.12.** The set of natural numbers: $\mathbb{N} = \{1, 2, 3, \ldots\}$. The set of real numbers: $\mathbb{R}$.

**Definition 1.1.13** (Equal Sets)**.** Two sets $A$ and $B$ are said to be equal if they have exactly the same elements. We write $A = B$.

**Example 1.1.14.** Let $A = \{1, 2, 3\}$ and $B = \{3, 1, 2\}$. Then $A = B$ because the order of elements in a set does not matter. Let $C = \{x \mid x^2 - 4x + 3 = 0\}$ and $D = \{1, 3\}$. Since the roots of $x^2 - 4x + 3 = 0$ are $x = 1$ and $x = 3$, $C = \{1, 3\}$. Thus, $C = D$.

**Definition 1.1.15** (Equivalent Sets)**.** Two finite sets $A$ and $B$ are said to be equivalent if they have the same number of elements. This is also called having the same cardinality. We write $A \sim B$ or $|A| = |B|$ or $n(A) = n(B)$.

**Example 1.1.16.** Let $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$. Then $A$ and $B$ are equivalent sets because $n(A) = 3$ and $n(B) = 3$. Note that equal sets are always equivalent, but equivalent sets are not necessarily equal.

### 1.1.3 Subsets

**Definition 1.1.17** (Subset)**.** A set $A$ is said to be a subset of a set $B$ if every element of $A$ is also an element of $B$. We write $A \subseteq B$. If $A \subseteq B$ and $A \neq B$, then $A$ is called a proper subset of $B$, denoted by $A \subset B$. If $A \subseteq B$, we can also say that $B$ is a superset of $A$, denoted by $B \supseteq A$.

**Theorem 1.1.18.**
1. *Every set is a subset of itself, i.e., $A \subseteq A$.*
2. *The empty set is a subset of every set, i.e., $\emptyset \subseteq A$.*

*Proof.*
1. To prove $A \subseteq A$, we need to show that every element of $A$ is also an element of $A$. This is trivially true by definition.
2. To prove $\emptyset \subseteq A$, we need to show that every element of $\emptyset$ is also an element of $A$. Since the empty set has no elements, the condition is vacuously true. There is no element in $\emptyset$ that is not in $A$.

$\blacksquare$

**Example 1.1.19.** Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4, 5\}$. Then $A \subseteq B$ because every element of $A$ (1, 2, 3) is also in $B$. Also, $A \subset B$ because $A \neq B$. Let $C = \{a, b\}$. The subsets of $C$ are $\emptyset, \{a\}, \{b\}, \{a, b\}$. The proper subsets of $C$ are $\emptyset, \{a\}, \{b\}$.

### 1.1.4 Basic Operations on Sets

**Definition 1.1.20** (Union)**.** The union of two sets $A$ and $B$, denoted by $A \cup B$, is the set of all elements that belong to $A$ or to $B$ (or to both). In set-builder form: $A \cup B = \{x \mid x \in A \text{ or } x \in B\}$.

**Example 1.1.21.** Let $A = \{1, 2, 3\}$ and $B = \{3, 4, 5\}$. Then $A \cup B = \{1, 2, 3, 4, 5\}$.

**Property 1.1.22** (Properties of Union)**.** *1. Commutative Law:* $A \cup B = B \cup A$

2. *Associative Law:* $(A \cup B) \cup C = A \cup (B \cup C)$
3. *Idempotent Law:* $A \cup A = A$
4. *Law of* $\emptyset$*:* $A \cup \emptyset = A$
5. *Law of* $U$ *(Universal Set):* $A \cup U = U$

**Definition 1.1.23** (Intersection)**.** The intersection of two sets $A$ and $B$, denoted by $A \cap B$, is the set of all elements that belong to both $A$ and $B$. In set-builder form: $A \cap B = \{x \mid x \in A \text{ and } x \in B\}$.

**Example 1.1.24.** Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. Then $A \cap B = \{3, 4\}$.

**Property 1.1.25** (Properties of Intersection)**.** *1. Commutative Law:* $A \cap B = B \cap A$

2. *Associative Law:* $(A \cap B) \cap C = A \cap (B \cap C)$
3. *Idempotent Law:* $A \cap A = A$
4. *Law of* $\emptyset$*:* $A \cap \emptyset = \emptyset$
5. *Law of* $U$ *(Universal Set):* $A \cap U = A$
6. *Distributive Law:* $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

**Definition 1.1.26** (Disjoint Sets)**.** Two sets $A$ and $B$ are said to be disjoint if their intersection is the empty set, i.e., $A \cap B = \emptyset$.

**Example 1.1.27.** Let $A = \{1, 2, 3\}$ and $B = \{4, 5, 6\}$. Then $A \cap B = \emptyset$, so $A$ and $B$ are disjoint sets.

**Definition 1.1.28** (Difference)**.** The difference of two sets $A$ and $B$, denoted by $A - B$, is the set of all elements that belong to $A$ but not to $B$. In set-builder form: $A - B = \{x \mid x \in A \text{ and } x \notin B\}$.

**Example 1.1.29.** Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 5, 6, 7\}$. Then $A - B = \{1, 2, 4\}$. Note that $B - A = \{6, 7\}$, which shows that $A - B \neq B - A$.

**Definition 1.1.30** (Complement of a Set)**.** Let $U$ be the universal set. The complement of a set $A$, denoted by $A'$ or $A^c$ or $\bar{A}$, is the set of all elements in $U$ that are not in $A$. In set-builder form: $A' = \{x \mid x \in U \text{ and } x \notin A\} = U - A$.

**Example 1.1.31.** Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ be the universal set. Let $A = \{2, 4, 6, 8, 10\}$ be the set of even numbers in $U$. Then $A' = \{1, 3, 5, 7, 9\}$.

**Property 1.1.32** (Properties of Complement)**.** *1. Complement Laws:* $A \cup A' = U$ $A \cap A' = \emptyset$

2. *Law of double complementation:* $(A')' = A$
3. *Laws of empty set and universal set:* $\emptyset' = U$ $U' = \emptyset$

**Definition 1.1.33** (Power Set)**.** The power set of a set $A$, denoted by $P(A)$, is the set of all possible subsets of $A$.

**Example 1.1.34.** Let $A = \{1, 2\}$. The subsets of $A$ are $\emptyset, \{1\}, \{2\}, \{1, 2\}$. Therefore, the power set of $A$ is $P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

**Theorem 1.1.35.** *If a set $A$ has $n$ elements, then its power set $P(A)$ has $2^n$ elements.*

*Proof.* Consider a set $A$ with $n$ elements. For each element in $A$, there are two possibilities when forming a subset: either the element is included in the subset or it is not. Since there are $n$ elements, and each has two independent choices, the total number of distinct subsets is $2 \times 2 \times \cdots \times 2$ (n times), which is $2^n$. ■

**Example 1.1.36.** Let $B = \{a, b, c\}$. Here, $n(B) = 3$. The number of elements in $P(B)$ should be $2^3 = 8$. $P(B) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$.

We previously defined finite and infinite sets. Let's delve a bit deeper with more formal definitions and examples.

**Definition 1.1.37** (Finite Set)**.** A set $A$ is said to be **finite** if it is either the empty set ($\emptyset$) or if it can be put into one-to-one correspondence with the set $\{1, 2, 3, \ldots, n\}$ for some natural number $n$. The number $n$ is called the cardinality of the set, denoted by $|A|$ or $n(A)$.

In simpler terms, a finite set has a fixed, non-negative integer number of elements that we can count and eventually finish.

**Example 1.1.38.**

1. The set of planets in our solar system: $P = \{$Mercury, Venus, Earth, Mars, Jupiter, S
   Here, $|P| = 8$. This is a finite set.
2. The set of prime numbers less than 20: $S = \{2, 3, 5, 7, 11, 13, 17, 19\}$. Here, $|S| = 8$. This is also a finite set.
3. The empty set $\emptyset$ is a finite set with cardinality $0$.

**Definition 1.1.39** (Infinite Set)**.** A set is said to be **infinite** if it is not finite. This means it cannot be put into one-to-one correspondence with any set $\{1, 2, \ldots, n\}$.

An infinite set has an unlimited number of elements. We cannot count all its elements to reach an end.

**Example 1.1.40.**

1. The set of natural numbers: $\mathbb{N} = \{1, 2, 3, \ldots\}$.
2. The set of integers: $\mathbb{Z} = \{\ldots, -2, -1, 0, 1, 2, \ldots\}$.
3. The set of real numbers: $\mathbb{R}$.
4. The set of points on a line segment.

### 1.1.5 Countable and Uncountable Sets

The concept of countability allows us to distinguish between different "sizes" of infinite sets.

**Definition 1.1.41** (Countable Set)**.** A set is said to be **countable** if it is either finite or can be put into one-to-one correspondence with the set of natural numbers $\mathbb{N}$. In other words, a set $A$ is countable if we can "list" its elements, possibly in an infinite sequence: $a_1, a_2, a_3, \ldots$.

**Definition 1.1.42** (Countably Infinite Set)**.** An infinite set that is countable is called a **countably infinite** set. This means it has the same cardinality as the set of natural numbers $\mathbb{N}$.

**Definition 1.1.43** (Uncountable Set)**.** A set is said to be **uncountable** if it is not countable. This means it is an infinite set that cannot be put into one-to-one correspondence with the set of natural numbers $\mathbb{N}$.

#### Examples of Countable Sets

**Example 1.1.44** (Natural Numbers)**.** The set of natural numbers $\mathbb{N} = \{1, 2, 3, \dots\}$ is countably infinite. The one-to-one correspondence is trivial: $f(n) = n$.

**Example 1.1.45** (Integers)**.** The set of integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$ is countably infinite. We can list its elements in a sequence: $0, 1, -1, 2, -2, 3, -3, \dots$. A formal one-to-one correspondence $f : \mathbb{N} \to \mathbb{Z}$ can be defined as: $f(n) = \begin{cases} (n-1)/2 & \text{if } n \text{ is odd} \\ -n/2 & \text{if } n \text{ is even} \end{cases}$

$$\begin{aligned} f(1) &= (1-1)/2 = 0 \\ f(2) &= -2/2 = -1 \\ f(3) &= (3-1)/2 = 1 \\ f(4) &= -4/2 = -2 \\ f(5) &= (5-1)/2 = 2 \\ &\dots \end{aligned}$$

This shows that $\mathbb{Z}$ has the same cardinality as $\mathbb{N}$.

**Example 1.1.46** (Rational Numbers)**.** The set of rational numbers $\mathbb{Q} = \{p/q \mid p \in \mathbb{Z}, q \in \mathbb{N}, \gcd(p, q) = 1\}$ is countably infinite. This can be shown using Cantor's diagonalization argument, but a simpler visualization is to arrange positive rationals in a grid and traverse them diagonally.

| $p/q$ | $q=1$ | $q=2$ | $q=3$ | $q=4$ | $q=5$ | $\dots$ |
|---|---|---|---|---|---|---|
| $p=1$ | $1/1$ | $1/2$ | $1/3$ | $1/4$ | $1/5$ | $\dots$ |
| $p=2$ | $2/1$ | $2/2$ | $2/3$ | $2/4$ | $2/5$ | $\dots$ |
| $p=3$ | $3/1$ | $3/2$ | $3/3$ | $3/4$ | $3/5$ | $\dots$ |
| $p=4$ | $4/1$ | $4/2$ | $4/3$ | $4/4$ | $4/5$ | $\dots$ |
| $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\vdots$ | $\ddots$ |

By listing them as $1/1, 1/2, 2/1, 1/3, 3/1, 1/4, 2/3, 3/2, 4/1, \dots$ (omitting duplicates like $2/2 = 1/1$), and then incorporating negative rationals and zero, we can create an infinite list, proving countability.

**Theorem 1.1.47.** *The union of a countable number of countable sets is countable.*

*Proof.* This can be proven by a similar diagonalization argument. If $A_1, A_2, \dots$ are countable sets, we can list the elements of each set and then combine these lists into a single countable list for their union. ■

#### Examples of Uncountable Sets

**Example 1.1.48** (Real Numbers)**.** The set of real numbers $\mathbb{R}$ is uncountable. This is a fundamental result proven by Cantor's diagonalization argument.

*Proof.* Assume, for contradiction, that the set of real numbers between 0 and 1 (denoted as $(0, 1)$) is countably infinite. This means we can list all such real numbers:

$$\begin{aligned} r_1 &= 0.d_{11}d_{12}d_{13}d_{14}\dots \\ r_2 &= 0.d_{21}d_{22}d_{23}d_{24}\dots \\ r_3 &= 0.d_{31}d_{32}d_{33}d_{34}\dots \\ &\dots \end{aligned}$$

Now, construct a new real number $r = 0.d_1d_2d_3d_4\ldots$ such that $d_i \neq d_{ii}$ (e.g., if $d_{ii} = 1$, choose $d_i = 2$; otherwise, choose $d_i = 1$). This new number $r$ is a real number between 0 and 1, but it differs from every $r_i$ in the list at the $i$-th decimal place. Therefore, $r$ cannot be in the list, contradicting our initial assumption that all real numbers between 0 and 1 were listed. Since the interval $(0, 1)$ is uncountable, and it is a subset of $\mathbb{R}$, $\mathbb{R}$ itself must be uncountable. ■

This implies that there are "more" real numbers than natural numbers, even though both sets are infinite.

**Example 1.1.49** (Irrational Numbers)**.** The set of irrational numbers $\mathbb{I} = \mathbb{R} \setminus \mathbb{Q}$ is uncountable.

*Proof.* Assume $\mathbb{I}$ is countable. Since $\mathbb{Q}$ is countable, and the union of two countable sets is countable, then $\mathbb{R} = \mathbb{Q} \cup \mathbb{I}$ would be countable. But we know $\mathbb{R}$ is uncountable. This is a contradiction, so $\mathbb{I}$ must be uncountable. ■

**Example 1.1.50** (Power Set of Natural Numbers)**.** The power set of the natural numbers $P(\mathbb{N})$ is uncountable. This can also be proven using Cantor's diagonalization argument. Each subset of $\mathbb{N}$ can be represented by an infinite binary sequence (where 1 means the element is in the subset, 0 means it's not). The set of all such infinite binary sequences is uncountable, as shown by a diagonalization argument similar to that for real numbers.

## 1.2 Cartesian Product

**Definition 1.2.1.** The **Cartesian product** of two sets $A$ and $B$, denoted $A \times B$, is the set of all ordered pairs $(a, b)$ where $a \in A$ and $b \in B$:

$$A \times B = \{(a, b) \mid a \in A \text{ and } b \in B\}$$

**Example 1.2.2.** 1. If $A = \{1, 2\}$ and $B = \{x, y\}$, then:

$$A \times B = \{(1, x), (1, y), (2, x), (2, y)\}$$

2. $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ represents the Cartesian plane.

**Theorem 1.2.3.** *For finite sets $A$ and $B$, $|A \times B| = |A| \cdot |B|$.*

**Example 1.2.4.** Let $U = \{1, 2, 3, 4, 5\}$, $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$. Then:

- $A \cup B = \{1, 2, 3, 4\}$
- $A \cap B = \{2, 3\}$
- $A^c = \{4, 5\}$
- $A \setminus B = \{1\}$

## 1.3 De Morgan's Laws

**Theorem 1.3.1** (De Morgan's Laws)**.** *For any sets $A$ and $B$:*

1. $(A \cup B)^c = A^c \cap B^c$
2. $(A \cap B)^c = A^c \cup B^c$

*Proof.* We prove the first law. Let $x \in (A \cup B)^c$. Then $x \notin A \cup B$, so $x \notin A$ and $x \notin B$. Thus $x \in A^c$ and $x \in B^c$, so $x \in A^c \cap B^c$.

Conversely, if $x \in A^c \cap B^c$, then $x \notin A$ and $x \notin B$, so $x \notin A \cup B$, hence $x \in (A \cup B)^c$.

The second law can be proved similarly. ■

## 1.4 Relations

**Definition 1.4.1.** A **relation** $R$ from set $A$ to set $B$ is a subset of $A \times B$. If $(a, b) \in R$, we write $aRb$.

**Example 1.4.2.** 1. The relation "less than" on $\mathbb{R}$: $R = \{(a, b) \in \mathbb{R} \times \mathbb{R} \mid a < b\}$

2. For $A = \{1, 2, 3\}$, the relation "divides": $R = \{(1, 1), (1, 2), (1, 3), (2, 2), (3, 3)\}$

**Definition 1.4.3.** A relation $R$ on a set $A$ is:

- **Reflexive** if $\forall a \in A, aRa$
- **Symmetric** if $\forall a, b \in A, aRb \Rightarrow bRa$
- **Transitive** if $\forall a, b, c \in A, aRb$ and $bRc \Rightarrow aRc$

## 1.5 Equivalence Relations

**Definition 1.5.1.** A relation $R$ on a set $A$ is an **equivalence relation** if it is reflexive, symmetric, and transitive.

**Example 1.5.2.** 1. Equality on any set is an equivalence relation.

2. Congruence modulo $n$ on $\mathbb{Z}$: $a \equiv b \pmod{n}$ if $n \mid (a - b)$
3. Similarity of triangles in geometry

**Definition 1.5.3.** Let $R$ be an equivalence relation on $A$. The **equivalence class** of $a \in A$ is:

$$[a] = \{b \in A \mid aRb\}$$

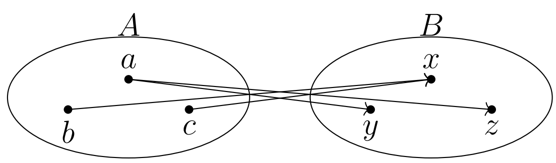
**Theorem 1.5.4.** *If $R$ is an equivalence relation on $A$, then:*

1. $\forall a \in A, a \in [a]$
2. $[a] = [b]$ *if and only if* $aRb$
3. *The equivalence classes partition $A$ (they are disjoint and their union is $A$)*

**Example 1.5.5.** For congruence modulo 3 on $\mathbb{Z}$, the equivalence classes are:

- $[0] = \{\ldots, -6, -3, 0, 3, 6, \ldots\}$
- $[1] = \{\ldots, -5, -2, 1, 4, 7, \ldots\}$
- $[2] = \{\ldots, -4, -1, 2, 5, 8, \ldots\}$

These three classes form a partition of $\mathbb{Z}$.




Cartesian Product

## 1.6 Introduction to Partial Orders

**Definition 1.6.1.** A **partially ordered set** (or **poset**) is a set $P$ equipped with a binary relation $\preceq$ that satisfies the following properties for all $a, b, c \in P$:

1. **Reflexivity**: $a \preceq a$
2. **Antisymmetry**: If $a \preceq b$ and $b \preceq a$, then $a = b$
3. **Transitivity**: If $a \preceq b$ and $b \preceq c$, then $a \preceq c$

The relation $\preceq$ is called a **partial order**.

**Example 1.6.2.**
1. The set of real numbers $\mathbb{R}$ with the usual ordering $\leq$ is a poset.
2. For any set $S$, the power set $\mathcal{P}(S)$ with the subset relation $\subseteq$ is a poset.
3. The set of natural numbers $\mathbb{N}$ with the divisibility relation ($a \preceq b$ if $a$ divides $b$) is a poset.
4. The set of all subspaces of a vector space, ordered by inclusion, forms a poset.

## 1.7 Comparability and Related Concepts

**Definition 1.7.1.** In a poset $(P, \preceq)$:

- Two elements $a, b \in P$ are **comparable** if either $a \preceq b$ or $b \preceq a$.
- If all pairs of elements are comparable, the poset is called a **totally ordered set** or **chain**.
- If no two distinct elements are comparable, the poset is called an **antichain**.

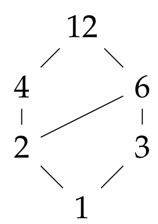
**Example 1.7.2.**
1. $(\mathbb{R}, \leq)$ is a chain.
2. For any set $S$ with more than one element, $(\mathcal{P}(S), \subseteq)$ is not a chain.
3. Any set with the equality relation ($a \preceq b$ iff $a = b$) is an antichain.

## 1.8 Hasse Diagrams

**Definition 1.8.1.** A **Hasse diagram** is a visual representation of a poset where:

- Elements are represented as points.
- If $a \preceq b$ and there is no $c$ such that $a \preceq c \preceq b$ (i.e., $b$ **covers** $a$), then we draw a line from $a$ to $b$ with $a$ placed below $b$.
- Lines implied by transitivity are omitted.

**Example 1.8.2.** The Hasse diagram for the poset $(\{1, 2, 3, 4, 6, 12\}, |)$ (divisibility relation) is:



## 1.9 Special Elements in Posets

**Definition 1.9.1.** Let $(P, \preceq)$ be a poset and let $S \subseteq P$.

- An element $a \in S$ is a **minimal element** of $S$ if there is no $b \in S$ with $b \prec a$.
- An element $a \in S$ is a **maximal element** of $S$ if there is no $b \in S$ with $a \prec b$.
- An element $a \in P$ is a **lower bound** for $S$ if $a \preceq s$ for all $s \in S$.
- An element $a \in P$ is an **upper bound** for $S$ if $s \preceq a$ for all $s \in S$.
- The **greatest lower bound** (or **meet**) of $S$ is the greatest element among all lower bounds of $S$.
- The **least upper bound** (or **join**) of $S$ is the least element among all upper bounds of $S$.

**Example 1.9.2.** In the poset $(\mathcal{P}(\{1, 2, 3\}), \subseteq)$:

- The minimal element is $\emptyset$.
- The maximal elements are all 1-element subsets: $\{1\}, \{2\}, \{3\}$.
- For $S = \{\{1\}, \{2\}\}$, the lower bounds are $\emptyset$ and the meet is $\emptyset$.
- For $S = \{\{1\}, \{2\}\}$, the upper bounds are $\{1, 2\}, \{1, 2, 3\}$ and the join is $\{1, 2\}$.

## 1.10 Lattices

**Definition 1.10.1.** A poset $(L, \preceq)$ is a **lattice** if every pair of elements $a, b \in L$ has both a meet (denoted $a \wedge b$) and a join (denoted $a \vee b$).

**Example 1.10.2.** 1. $(\mathbb{R}, \leq)$ is a lattice with $a \wedge b = \min(a, b)$ and $a \vee b = \max(a, b)$.

2. $(\mathcal{P}(S), \subseteq)$ is a lattice with $A \wedge B = A \cap B$ and $A \vee B = A \cup B$.
3. The set of natural numbers with the divisibility relation is a lattice with $a \wedge b = \gcd(a, b)$ and $a \vee b = \operatorname{lcm}(a, b)$.

## 1.11 Complete Lattices

**Definition 1.11.1.** A lattice $L$ is **complete** if every subset $S \subseteq L$ has both a meet and a join.

**Theorem 1.11.2.** *A lattice $L$ is complete if and only if every subset of $L$ has a meet (or equivalently, every subset has a join).*

**Example 1.11.3.** 1. $(\mathcal{P}(S), \subseteq)$ is a complete lattice.

2. The closed interval $[0, 1]$ with the usual order is a complete lattice.
3. $(\mathbb{N}, |)$ (divisibility) is not complete (consider the set of all primes).

## 1.12 Applications of Posets

- **Computer Science**: Used in domain theory, formal concept analysis, and fixed point theorems.
- **Mathematics**: Fundamental in algebra (lattice theory), combinatorics, and topology.
- **Physics**: Used in quantum mechanics (projection lattices) and statistical mechanics.
- **Data Analysis**: Concept lattices in formal concept analysis.

## Summary

Partial orders provide a fundamental framework for understanding hierarchical relationships in mathematics and computer science. The concepts of posets, lattices, and their special elements form the basis for more advanced topics in order theory and its applications.

# Chapter 2

# Real and Complex Number Systems

## 2.1 The Real Number System

**Definition 2.1.1.** The **real number system**, denoted by $\mathbb{R}$, is the complete ordered field that contains all rational and irrational numbers. It can be visualized as all points on an infinitely long number line.

The real numbers include:

- Natural numbers: $\mathbb{N} = \{1, 2, 3, \dots\}$
- Whole numbers: $\{0, 1, 2, 3, \dots\}$
- Integers: $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- Rational numbers: $\mathbb{Q} = \{\frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0\}$
- Irrational numbers: Numbers that cannot be expressed as $\frac{p}{q}$ where $p, q \in \mathbb{Z}$

## 2.2 Rational Numbers

**Definition 2.2.1.** A number $r$ is called a **rational number** if it can be expressed in the form $r = \frac{p}{q}$, where $p$ and $q$ are integers and $q \neq 0$.

**Example 2.2.2.** 1. $\frac{1}{2}$, $\frac{3}{4}$, $\frac{-7}{5}$ are rational numbers.

2. All integers are rational numbers (e.g., $5 = \frac{5}{1}$).
3. Terminating decimals: $0.25 = \frac{1}{4}$
4. Repeating decimals: $0.\overline{3} = \frac{1}{3}$, $0.\overline{142857} = \frac{1}{7}$

**Property 1** (Properties of Rational Numbers)**.** *1. The set $\mathbb{Q}$ is countable.*

2. *Between any two rational numbers, there exists another rational number.*
3. *Rational numbers are dense in $\mathbb{R}$.*
4. *The decimal expansion of a rational number is either terminating or repeating.*

## 2.3 Irrational Numbers

**Definition 2.3.1.** A number is called an **irrational number** if it cannot be expressed as a ratio of two integers. Its decimal expansion is non-terminating and non-repeating.

**Example 2.3.2.** 1. $\sqrt{2}$ (proved irrational by the ancient Greeks)

2. $\pi$ (ratio of circumference to diameter of a circle)
3. $e$ (base of natural logarithms)
4. $\phi$ (golden ratio, $\frac{1+\sqrt{5}}{2}$)

**Theorem 2.3.3.** *$\sqrt{2}$ is irrational.*

*Proof.* Assume $\sqrt{2}$ is rational. Then $\sqrt{2} = \frac{p}{q}$ where $p$ and $q$ are coprime integers. Then $2 = \frac{p^2}{q^2}$, so $p^2 = 2q^2$. This implies $p^2$ is even, so $p$ is even. Let $p = 2k$. Then $4k^2 = 2q^2$, so $q^2 = 2k^2$, which means $q^2$ is even, so $q$ is even. But this contradicts that $p$ and $q$ are coprime. Therefore, $\sqrt{2}$ is irrational. ■

## 2.4 Closure Properties of Real Numbers

**Definition 2.4.1.** A set is said to be **closed** under an operation if performing that operation on members of the set always produces a member of the same set.

**Property 2** (Closure Properties of $\mathbb{R}$). *The set of real numbers is closed under the following operations:*

1. ***Addition**: If $a, b \in \mathbb{R}$, then $a + b \in \mathbb{R}$*
2. ***Subtraction**: If $a, b \in \mathbb{R}$, then $a - b \in \mathbb{R}$*
3. ***Multiplication**: If $a, b \in \mathbb{R}$, then $a \cdot b \in \mathbb{R}$*
4. ***Division**: If $a, b \in \mathbb{R}$ and $b \neq 0$, then $\frac{a}{b} \in \mathbb{R}$*

**Example 2.4.2.** 1. $2 + 3 = 5 \in \mathbb{R}$

2. $5 - 7 = -2 \in \mathbb{R}$
3. $4 \times 6 = 24 \in \mathbb{R}$
4. $\frac{8}{2} = 4 \in \mathbb{R}$, $\frac{1}{3} \in \mathbb{R}$

**Property 3** (Non-Closure Properties). *1. The set of irrational numbers is **not closed** under addition or multiplication.*

2. *Example: $\sqrt{2} + (-\sqrt{2}) = 0$ (rational)*
3. *Example: $\sqrt{2} \times \sqrt{2} = 2$ (rational)*

## 2.5 Completeness Property of Real Numbers

**Theorem 2.5.1** (Completeness Axiom). *Every non-empty set of real numbers that is bounded above has a least upper bound (supremum) in $\mathbb{R}$.*

This property distinguishes $\mathbb{R}$ from $\mathbb{Q}$ and is fundamental to calculus and real analysis.

**Example 2.5.2.** The set $\{x \in \mathbb{Q} \mid x^2 < 2\}$ is bounded above in $\mathbb{Q}$ but has no least upper bound in $\mathbb{Q}$. However, in $\mathbb{R}$, its least upper bound is $\sqrt{2}$.

## Summary

The real number system forms a complete ordered field that includes both rational and irrational numbers. While rational numbers can be expressed as fractions of integers, irrational numbers have non-terminating, non-repeating decimal expansions. The closure properties of real numbers under basic arithmetic operations make them fundamental to mathematics, and the completeness property distinguishes them from other number systems.

## 2.6 Introduction to Complex Numbers

**Definition 2.6.1.** A **complex number** is a number of the form $z = a + bi$, where:

- $a$ and $b$ are real numbers
- $i$ is the imaginary unit with the property $i^2 = -1$
- $a$ is called the **real part**, denoted $\mathrm{Re}(z)$
- $b$ is called the **imaginary part**, denoted $\mathrm{Im}(z)$

The set of all complex numbers is denoted by $\mathbb{C}$.

**Example 2.6.2.** 1. $3 + 4i$ (Real part = 3, Imaginary part = 4)

2. $-2 - i$ (Real part = -2, Imaginary part = -1)
3. $5$ (Real part = 5, Imaginary part = 0)
4. $2i$ (Real part = 0, Imaginary part = 2)

## 2.7 Equality of Complex Numbers

**Definition 2.7.1.** Two complex numbers $z_1 = a + bi$ and $z_2 = c + di$ are **equal** if and only if:

$$a = c \quad \text{and} \quad b = d$$

That is, their real parts are equal and their imaginary parts are equal.

**Example 2.7.2.** 1. $2 + 3i = 2 + 3i$

2. $x + yi = 3 - 2i$ if and only if $x = 3$ and $y = -2$
3. $4 + 2i \neq 4 - 2i$ (different imaginary parts)
4. $5 + 0i = 5$ (a real number)

## 2.8 Operations on Complex Numbers

### 2.8.1 Addition and Subtraction

**Definition 2.8.1.** For complex numbers $z_1 = a + bi$ and $z_2 = c + di$:

- **Addition**: $z_1 + z_2 = (a + c) + (b + d)i$
- **Subtraction**: $z_1 - z_2 = (a - c) + (b - d)i$

**Example 2.8.2.** Let $z_1 = 3 + 2i$ and $z_2 = 1 - 4i$. Then:

$$z_1 + z_2 = (3 + 1) + (2 + (-4))i = 4 - 2i$$
$$z_1 - z_2 = (3 - 1) + (2 - (-4))i = 2 + 6i$$

### 2.8.2 Multiplication

**Definition 2.8.3.** For complex numbers $z_1 = a + bi$ and $z_2 = c + di$:

$$z_1 \cdot z_2 = (ac - bd) + (ad + bc)i$$

This is obtained by distributing and using $i^2 = -1$.

**Example 2.8.4.** Let $z_1 = 2 + 3i$ and $z_2 = 1 - 2i$. Then:

$$\begin{aligned} z_1 \cdot z_2 &= (2)(1) + (2)(-2i) + (3i)(1) + (3i)(-2i) \\ &= 2 - 4i + 3i - 6i^2 \\ &= 2 - i - 6(-1) \\ &= 2 - i + 6 = 8 - i \end{aligned}$$

### 2.8.3 Complex Conjugate

**Definition 2.8.5.** The **complex conjugate** of $z = a + bi$ is denoted $\overline{z}$ and defined as:

$$\overline{z} = a - bi$$

**Example 2.8.6.** 1. If $z = 3 + 4i$, then $\overline{z} = 3 - 4i$

2. If $z = 2 - i$, then $\overline{z} = 2 + i$
3. If $z = 5$ (real), then $\overline{z} = 5$
4. If $z = 3i$ (purely imaginary), then $\overline{z} = -3i$

### 2.8.4 Division

**Definition 2.8.7.** For complex numbers $z_1 = a + bi$ and $z_2 = c + di \neq 0$:

$$\frac{z_1}{z_2} = \frac{a + bi}{c + di} = \frac{(a + bi)(c - di)}{(c + di)(c - di)} = \frac{(ac + bd) + (bc - ad)i}{c^2 + d^2}$$

We multiply numerator and denominator by the conjugate of the denominator.

**Example 2.8.8.** Let $z_1 = 3 + 2i$ and $z_2 = 1 - i$. Then:

$$\begin{aligned} \frac{z_1}{z_2} &= \frac{3 + 2i}{1 - i} = \frac{(3 + 2i)(1 + i)}{(1 - i)(1 + i)} \\ &= \frac{3 + 3i + 2i + 2i^2}{1 - i^2} = \frac{3 + 5i - 2}{1 + 1} \\ &= \frac{1 + 5i}{2} = \frac{1}{2} + \frac{5}{2}i \end{aligned}$$

### 2.8.5 Modulus (Absolute Value)

**Definition 2.8.9.** The **modulus** (or absolute value) of $z = a + bi$ is denoted $|z|$ and defined as:

$$|z| = \sqrt{a^2 + b^2}$$

**Example 2.8.10.** 1. $|3 + 4i| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$

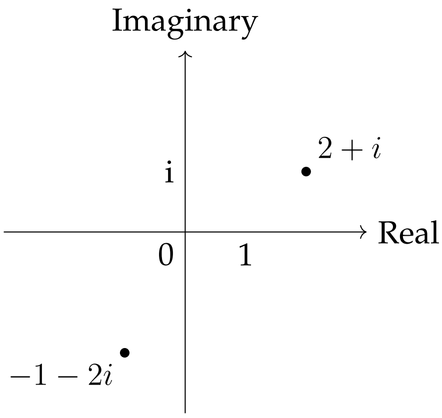
2. $|1 - i| = \sqrt{1^2 + (-1)^2} = \sqrt{1 + 1} = \sqrt{2}$
3. $|5| = \sqrt{5^2 + 0^2} = 5$
4. $|2i| = \sqrt{0^2 + 2^2} = 2$

## 2.9 Geometric Representation

Complex numbers can be represented as points in the **complex plane** (also called the Argand diagram), where:

- The x-axis represents the real part
- The y-axis represents the imaginary part




## 2.10 Properties of Complex Operations

**Theorem 2.10.1.** *For any complex numbers $z, z_1, z_2, z_3 \in \mathbb{C}$:*

1. ***Commutativity***: $z_1 + z_2 = z_2 + z_1$, $z_1 \cdot z_2 = z_2 \cdot z_1$
2. ***Associativity***: $(z_1 + z_2) + z_3 = z_1 + (z_2 + z_3)$, $(z_1 \cdot z_2) \cdot z_3 = z_1 \cdot (z_2 \cdot z_3)$
3. ***Distributivity***: $z_1 \cdot (z_2 + z_3) = z_1 \cdot z_2 + z_1 \cdot z_3$
4. ***Conjugate properties:*** $\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$, $\overline{z_1 \cdot z_2} = \overline{z_1} \cdot \overline{z_2}$
5. ***Modulus properties:*** $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$, $\left|\frac{z_1}{z_2}\right| = \frac{|z_1|}{|z_2|}$ *for* $z_2 \neq 0$

# Chapter 3

# Coordinate System

## 3.1 The Rectangular Coordinate System

**Definition 3.1.1.** The **rectangular coordinate system** (or **Cartesian coordinate system**) is a method of representing points in a plane using two perpendicular number lines, called the **axes**. The horizontal axis is the **x-axis**, and the vertical axis is the **y-axis**. Their intersection is called the **origin**, denoted by $O(0,0)$. A point $P$ in the plane is represented by an ordered pair of real numbers $(x, y)$, where $x$ is the **x-coordinate** (abscissa) and $y$ is the **y-coordinate** (ordinate).

### 3.1.1 Distance Formula

**Theorem 3.1.2** (Distance Formula)**.** *The distance between two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ in a rectangular coordinate system is given by the formula:*

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

*Proof.* Consider the two points $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$. We can construct a right-angled triangle with the hypotenuse as the segment connecting $P_1$ and $P_2$. Let the third vertex of this triangle be $Q(x_2, y_1)$.

The length of the horizontal side, $P_1Q$, is the absolute difference in the x-coordinates: $|x_2 - x_1|$. The length of the vertical side, $P_2Q$, is the absolute difference in the y-coordinates: $|y_2 - y_1|$.

By the Pythagorean theorem, the square of the hypotenuse is equal to the sum of the squares of the other two sides:

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

Taking the square root of both sides gives the distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

■

**Example 3.1.3.** Find the distance between the points $A(3, -2)$ and $B(-1, 5)$.

*Solution.* Let $(x_1, y_1) = (3, -2)$ and $(x_2, y_2) = (-1, 5)$. Using the distance formula:

$$d = \sqrt{(-1 - 3)^2 + (5 - (-2))^2}$$

$$d = \sqrt{(-4)^2 + (5 + 2)^2}$$

$$d = \sqrt{16 + 7^2}$$
$$d = \sqrt{16 + 49}$$
$$d = \sqrt{65}$$

The distance between the points is $\sqrt{65}$ units. ■

### 3.1.2 Section Formulae

**Theorem 3.1.4** (Internal Section Formula)**.** *The coordinates of a point $P(x, y)$ that divides the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the ratio $m : n$ are given by:*

$$x = \frac{mx_2 + nx_1}{m + n}, \quad y = \frac{my_2 + ny_1}{m + n}$$

*Proof.* Let the point $P(x, y)$ divide the line segment $AB$ in the ratio $m : n$, so $\frac{AP}{PB} = \frac{m}{n}$.

Construct two perpendicular lines from $A$, $P$, and $B$ to the x-axis, meeting it at $A'$, $P'$, and $B'$ respectively. From the similarity of triangles, $\triangle APA''$ and $\triangle PBB''$ (where $A''$ and $B''$ are points on the perpendiculars), we have:

$$\frac{AP}{PB} = \frac{A'P'}{P'B'}$$

The lengths of the segments on the x-axis are: $A'P' = x - x_1$ $P'B' = x_2 - x$ Substituting these into the ratio:

$$\frac{m}{n} = \frac{x - x_1}{x_2 - x}$$
$$m(x_2 - x) = n(x - x_1)$$
$$mx_2 - mx = nx - nx_1$$
$$mx_2 + nx_1 = nx + mx$$
$$mx_2 + nx_1 = x(m + n)$$
$$x = \frac{mx_2 + nx_1}{m + n}$$

A similar proof can be used for the y-coordinate. ■

**Example 3.1.5.** Find the coordinates of the point that divides the line segment joining $A(1, 3)$ and $B(4, 7)$ in the ratio $2 : 3$ internally.

*Solution.* Here, $(x_1, y_1) = (1, 3)$, $(x_2, y_2) = (4, 7)$, $m = 2$, and $n = 3$. Using the internal section formula:

$$x = \frac{2(4) + 3(1)}{2 + 3} = \frac{8 + 3}{5} = \frac{11}{5}$$
$$y = \frac{2(7) + 3(3)}{2 + 3} = \frac{14 + 9}{5} = \frac{23}{5}$$

The coordinates of the point are $(\frac{11}{5}, \frac{23}{5})$. ■

**Theorem 3.1.6** (Midpoint Formula)**.** *The coordinates of the midpoint of a line segment joining $A(x_1, y_1)$ and $B(x_2, y_2)$ are:*

$$x = \frac{x_1 + x_2}{2}, \quad y = \frac{y_1 + y_2}{2}$$

*Proof.* The midpoint is a special case of the internal section formula where the ratio is $1 : 1$. Substituting $m = 1$ and $n = 1$ into the internal section formula, we get:

$$x = \frac{1 \cdot x_2 + 1 \cdot x_1}{1+1} = \frac{x_1 + x_2}{2}$$

$$y = \frac{1 \cdot y_2 + 1 \cdot y_1}{1+1} = \frac{y_1 + y_2}{2}$$

∎

**Theorem 3.1.7** (External Section Formula). *The coordinates of a point $P(x, y)$ that divides the line segment joining the points $A(x_1, y_1)$ and $B(x_2, y_2)$ externally in the ratio $m : n$ are given by:*

$$x = \frac{mx_2 - nx_1}{m-n}, \quad y = \frac{my_2 - ny_1}{m-n}$$

*Proof.* The proof is similar to that of the internal section formula, using similar triangles. The key difference is that the point $P$ lies outside the segment $AB$, which introduces the subtraction in the numerator and denominator. ∎

**Example 3.1.8.** Find the coordinates of the point that divides the line segment joining $A(2, 5)$ and $B(8, 11)$ in the ratio $4 : 1$ externally.

*Solution.* Here, $(x_1, y_1) = (2, 5)$, $(x_2, y_2) = (8, 11)$, $m = 4$, and $n = 1$. Using the external section formula:

$$x = \frac{4(8) - 1(2)}{4-1} = \frac{32-2}{3} = \frac{30}{3} = 10$$

$$y = \frac{4(11) - 1(5)}{4-1} = \frac{44-5}{3} = \frac{39}{3} = 13$$

The coordinates of the point are $(10, 13)$. ∎

## 3.2 Equation of a Straight Line

### 3.2.1 Various Forms of the Equation of a Line

**Definition 3.2.1.** The **slope** (or **gradient**) of a line is a measure of its steepness. It is the ratio of the change in the y-coordinate to the change in the x-coordinate between any two distinct points on the line. The slope, denoted by $m$, for a line passing through points $(x_1, y_1)$ and $(x_2, y_2)$ is given by:

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

**Definition 3.2.2.** The **y-intercept** of a line is the y-coordinate of the point where the line crosses the y-axis. It is the value of $y$ when $x = 0$.

**Theorem 3.2.3** (Slope-Intercept Form). *The equation of a line with slope $m$ and y-intercept $c$ is given by:*

$$y = mx + c$$

**Example 3.2.4.** Find the equation of the line with a slope of 3 and a y-intercept of -2.

*Solution.* Given $m = 3$ and $c = -2$. Using the slope-intercept form $y = mx + c$, we get:

$$y = 3x - 2$$

∎

**Theorem 3.2.5** (Point-Slope Form). *The equation of a line passing through a point $(x_1, y_1)$ with a slope $m$ is given by:*

$$y - y_1 = m(x - x_1)$$

*Proof.* Let $(x, y)$ be any arbitrary point on the line. Using the definition of slope with points $(x, y)$ and $(x_1, y_1)$, we have:

$$m = \frac{y - y_1}{x - x_1}$$

Multiplying both sides by $(x - x_1)$ gives:

$$y - y_1 = m(x - x_1)$$

■

**Example 3.2.6.** Find the equation of the line that passes through the point $(4, 1)$ and has a slope of -2.

*Solution.* Given $(x_1, y_1) = (4, 1)$ and $m = -2$. Using the point-slope form $y - y_1 = m(x - x_1)$:

$$y - 1 = -2(x - 4)$$

$$y - 1 = -2x + 8$$

$$y = -2x + 9$$

■

**Theorem 3.2.7** (Two-Point Form). *The equation of a line passing through two distinct points $(x_1, y_1)$ and $(x_2, y_2)$ is given by:*

$$y - y_1 = \left(\frac{y_2 - y_1}{x_2 - x_1}\right)(x - x_1)$$

*Proof.* This form is derived directly from the point-slope form by replacing the slope $m$ with its definition using two points, $m = \frac{y_2 - y_1}{x_2 - x_1}$. ■

**Example 3.2.8.** Find the equation of the line passing through the points $(1, -3)$ and $(5, 5)$.

*Solution.* Let $(x_1, y_1) = (1, -3)$ and $(x_2, y_2) = (5, 5)$. The slope is $m = \frac{5-(-3)}{5-1} = \frac{8}{4} = 2$. Using the point-slope form with the point $(1, -3)$:

$$y - (-3) = 2(x - 1)$$

$$y + 3 = 2x - 2$$

$$y = 2x - 5$$

■

**Theorem 3.2.9** (Intercept Form). *The equation of a line with x-intercept $a$ and y-intercept $b$ is given by:*

$$\frac{x}{a} + \frac{y}{b} = 1, \quad a, b \neq 0$$

### 3.2.2 Angle Between Two Lines

**Theorem 3.2.10.** *The angle $\theta$ between two non-vertical lines with slopes $m_1$ and $m_2$ is given by:*

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$$

*Proof.* Let the two lines be $L_1$ and $L_2$ with slopes $m_1 = \tan\alpha_1$ and $m_2 = \tan\alpha_2$, where $\alpha_1$ and $\alpha_2$ are the angles the lines make with the positive x-axis. The angle between the lines, $\theta$, is the difference between these angles, i.e., $\theta = |\alpha_2 - \alpha_1|$. Using the tangent of the difference of two angles formula, $\tan(\alpha_2 - \alpha_1) = \frac{\tan\alpha_2 - \tan\alpha_1}{1+\tan\alpha_2\tan\alpha_1}$, we get:

$$\tan\theta = \left|\frac{m_2 - m_1}{1 + m_1 m_2}\right|$$

The absolute value is used because the angle between two lines is conventionally taken as the acute angle. ■

**Theorem 3.2.11** (Parallel Lines)**.** *Two non-vertical lines are **parallel** if and only if their slopes are equal.*

$$m_1 = m_2$$

*Proof.* If two lines are parallel, they make the same angle with the x-axis, so $\alpha_1 = \alpha_2$. Therefore, $\tan\alpha_1 = \tan\alpha_2$, which means $m_1 = m_2$. ■

**Theorem 3.2.12** (Perpendicular Lines)**.** *Two non-vertical lines are **perpendicular** if and only if the product of their slopes is -1.*

$$m_1 m_2 = -1 \quad \textit{or} \quad m_2 = -\frac{1}{m_1}$$

*Proof.* If two lines are perpendicular, the angle between them is $\theta = 90°$. The formula for the angle between two lines gives $\tan 90° = \frac{m_2 - m_1}{1+m_1 m_2}$. Since $\tan 90°$ is undefined, the denominator must be zero:

$$1 + m_1 m_2 = 0$$

$$m_1 m_2 = -1$$

■

**Example 3.2.13.** Find the angle between the lines $y = 2x + 1$ and $3x + y = 5$.

*Solution.* The first line is in slope-intercept form, so its slope is $m_1 = 2$. The second line, $3x + y = 5$, can be rewritten as $y = -3x + 5$. Its slope is $m_2 = -3$. Using the formula for the angle between two lines:

$$\tan\theta = \left|\frac{-3 - 2}{1 + (2)(-3)}\right| = \left|\frac{-5}{1 - 6}\right| = \left|\frac{-5}{-5}\right| = 1$$

Since $\tan\theta = 1$, the angle $\theta = \arctan(1) = 45°$. ■

## 3.3 Second Degree Homogeneous Equations

**Definition 3.3.1.** A **second degree homogeneous equation** in two variables $x$ and $y$ is an equation of the form:

$$ax^2 + 2hxy + by^2 = 0$$

where $a$, $h$, and $b$ are real constants, not all zero.

This equation represents a pair of straight lines passing through the origin $(0, 0)$.

## 3.4 Condition for Representing Straight Lines

**Theorem 3.4.1.** *The equation $ax^2 + 2hxy + by^2 = 0$ represents a pair of straight lines (real and distinct, real and coincident, or imaginary) if and only if:*

$$h^2 - ab \geq 0$$

*Specifically:*

- *If $h^2 - ab > 0$: Two distinct real lines*
- *If $h^2 - ab = 0$: Two coincident lines (a single line)*
- *If $h^2 - ab < 0$: Two imaginary lines*

## 3.5 Finding the Individual Lines

To find the equations of the individual lines, we can factor the homogeneous equation or solve for $y$ in terms of $x$:

$$by^2 + 2hxy + ax^2 = 0$$

Treating this as a quadratic in $y$:

$$y = \frac{-2hx \pm \sqrt{(2hx)^2 - 4b(ax^2)}}{2b} = \frac{-2hx \pm 2x\sqrt{h^2 - ab}}{2b} = \frac{-h \pm \sqrt{h^2 - ab}}{b}x$$

Thus, the two lines are:

$$L_1 : y = m_1 x \quad \text{and} \quad L_2 : y = m_2 x$$

where

$$m_1 = \frac{-h + \sqrt{h^2 - ab}}{b}, \quad m_2 = \frac{-h - \sqrt{h^2 - ab}}{b}$$

## 3.6 Angle Between the Lines

**Theorem 3.6.1.** *The angle $\theta$ between the lines represented by $ax^2 + 2hxy + by^2 = 0$ is given by:*

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{|a + b|}$$

*If $a + b = 0$, then the lines are perpendicular ($\theta = 90°$).*

*Proof.* Let the slopes of the lines be $m_1$ and $m_2$. The angle between them is given by:

$$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1 m_2}\right|$$

From the quadratic in $m$ (obtained by dividing the original equation by $x^2$):

$$bm^2 + 2hm + a = 0$$

we have:

$$m_1 + m_2 = -\frac{2h}{b}, \quad m_1 m_2 = \frac{a}{b}$$

Then:

$$(m_1 - m_2)^2 = (m_1 + m_2)^2 - 4m_1m_2 = \frac{4h^2}{b^2} - \frac{4a}{b} = \frac{4(h^2 - ab)}{b^2}$$

$$|m_1 - m_2| = \frac{2\sqrt{h^2 - ab}}{|b|}$$

Also:

$$1 + m_1m_2 = 1 + \frac{a}{b} = \frac{a + b}{b}$$

Therefore:

$$\tan\theta = \frac{\frac{2\sqrt{h^2-ab}}{|b|}}{\left|\frac{a+b}{b}\right|} = \frac{2\sqrt{h^2 - ab}}{|a + b|}$$

If $a + b = 0$, then $1 + m_1m_2 = 0$, which implies $m_1m_2 = -1$, meaning the lines are perpendicular. ∎

## 3.7 Special Cases

1. **Perpendicular Lines**: The lines are perpendicular if $a + b = 0$.
2. **Coincident Lines**: The lines coincide if $h^2 = ab$.
3. **Imaginary Lines**: The lines are imaginary if $h^2 < ab$.

## 3.8 Examples

**Example 3.8.1.** Find the lines represented by $x^2 - 5xy + 6y^2 = 0$ and the angle between them.

Here, $a = 1$, $h = -\frac{5}{2}$, $b = 6$.

The condition: $h^2 - ab = \frac{25}{4} - 6 = \frac{1}{4} > 0$, so real distinct lines.

The lines are:

$$y = \frac{\frac{5}{2} \pm \frac{1}{2}}{6}x \Rightarrow y = \frac{1}{2}x \quad \text{and} \quad y = \frac{1}{3}x$$

Angle between them:

$$\tan\theta = \frac{2\sqrt{\frac{1}{4}}}{|1 + 6|} = \frac{2 \cdot \frac{1}{2}}{7} = \frac{1}{7}$$

$$\theta = \tan^{-1}\left(\frac{1}{7}\right)$$

**Example 3.8.2.** Show that $x^2 + 4xy + y^2 = 0$ represents two perpendicular lines.

Here, $a = 1$, $h = 2$, $b = 1$. Since $a + b = 2 \neq 0$, we check:

$$\tan\theta = \frac{2\sqrt{4 - 1}}{|1 + 1|} = \frac{2\sqrt{3}}{2} = \sqrt{3} \Rightarrow \theta = 60^\circ$$

Wait, this doesn't give perpendicular lines. Let me check: Actually, for perpendicular lines, we need $a + b = 0$. Here $1 + 1 = 2 \neq 0$. So these are not perpendicular.

For perpendicular lines, consider $x^2 + 2xy - y^2 = 0$. Here $a = 1$, $b = -1$, so $a + b = 0$, and indeed the lines are perpendicular.

## 3.9 Introduction to Matrices

A **matrix** is a rectangular array of numbers, symbols, or expressions arranged in rows and columns. It's a fundamental concept in linear algebra used to represent linear transformations, systems of linear equations, and more. A matrix with $m$ rows and $n$ columns is called an $m \times n$ matrix.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

The element in the $i^{th}$ row and $j^{th}$ column is denoted by $a_{ij}$.

## 3.10 Types of Matrices

### 3.10.1 Row Matrix

A matrix with only one row. Example: $A = \begin{pmatrix} 1 & 5 & 9 \end{pmatrix}$

### 3.10.2 Column Matrix

A matrix with only one column. Example: $B = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$

### 3.10.3 Square Matrix

A matrix with an equal number of rows and columns ($m = n$). Example: $C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

### 3.10.4 Diagonal Matrix

A square matrix where all the non-diagonal elements are zero. The main diagonal elements can be non-zero. Example: $D = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 8 \end{pmatrix}$

### 3.10.5 Identity Matrix

A special type of diagonal matrix where all the main diagonal elements are 1 and all other elements are 0. It is denoted by $I_n$, where $n$ is the order of the matrix. Example: $I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

### 3.10.6 Zero Matrix (or Null Matrix)

A matrix where all elements are zero. It is denoted by $O$. Example: $O_{2\times3} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

### 3.10.7 Symmetric Matrix

A square matrix $A$ is symmetric if $A^T = A$, where $A^T$ is the transpose of $A$. This means $a_{ij} = a_{ji}$ for all $i, j$. Example: $E = \begin{pmatrix} 1 & 7 & 3 \\ 7 & 4 & 5 \\ 3 & 5 & 6 \end{pmatrix}$

### 3.10.8 Skew-Symmetric Matrix

A square matrix $A$ is skew-symmetric if $A^T = -A$. This implies $a_{ij} = -a_{ji}$ and the main diagonal elements are all zero ($a_{ii} = 0$). Example: $F = \begin{pmatrix} 0 & 2 & 1 \\ -2 & 0 & 3 \\ -1 & -3 & 0 \end{pmatrix}$

## 3.11 Algebra of Matrices

### 3.11.1 Addition and Subtraction

Two matrices can be added or subtracted only if they have the same dimensions ($m \times n$). The operation is performed element-wise. If $A = (a_{ij})$ and $B = (b_{ij})$, then $A \pm B = (a_{ij} \pm b_{ij})$.

### 3.11.2 Scalar Multiplication

To multiply a matrix by a scalar $k$, multiply every element of the matrix by that scalar. If $A = (a_{ij})$, then $kA = (ka_{ij})$.

### 3.11.3 Matrix Multiplication

Matrix multiplication $AB$ is defined only if the number of columns of matrix $A$ is equal to the number of rows of matrix $B$. If $A$ is an $m \times n$ matrix and $B$ is an $n \times p$ matrix, their product $C = AB$ is an $m \times p$ matrix. The element $c_{ij}$ of the product matrix $C$ is the sum of the products of the elements of the $i^{th}$ row of $A$ and the corresponding elements of the $j^{th}$ column of $B$.

$$c_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

Matrix multiplication is generally not commutative, i.e., $AB \neq BA$.

### 3.11.4 Transpose of a Matrix

The transpose of a matrix $A$, denoted by $A^T$, is obtained by interchanging its rows and columns. If $A = (a_{ij})$ is an $m \times n$ matrix, then $A^T = (a_{ji})$ is an $n \times m$ matrix. Properties: $(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(kA)^T = kA^T$, $(AB)^T = B^T A^T$.

## 3.12 Determinant of a Square Matrix

The **determinant** is a scalar value that can be computed from the elements of a square matrix. It provides important properties of the matrix, such as its invertibility. The determinant of a matrix $A$ is denoted as $\det(A)$ or $|A|$.

### 3.12.1 Determinant of a $1 \times 1$ Matrix

For $A = (a_{11})$, $|A| = a_{11}$.

### 3.12.2 Determinant of a $2 \times 2$ Matrix

For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $|A| = ad - bc$.

### 3.12.3 Determinant of a $3 \times 3$ Matrix (Sarrus' Rule)

For $A = \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$, $|A| = a(ei - fh) - b(di - fg) + c(dh - eg)$. This can be expanded along any row or column using cofactors. The minor $M_{ij}$ of element $a_{ij}$ is the determinant of the submatrix obtained by deleting the $i^{th}$ row and $j^{th}$ column. The cofactor $C_{ij}$ is given by $C_{ij} = (-1)^{i+j} M_{ij}$. Then, the determinant is the sum of the products of the elements of any row or column with their corresponding cofactors. For the first row: $|A| = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$.

### 3.12.4 Properties of Determinants

- The determinant of a matrix and its transpose are equal: $|A| = |A^T|$.
- If any two rows or columns of a matrix are interchanged, the sign of the determinant changes.
- If any two rows or columns of a matrix are identical, the determinant is zero.
- If a matrix has a row or column of all zeros, its determinant is zero.
- The determinant of a product of two matrices is the product of their determinants: $|AB| = |A||B|$.
- A square matrix $A$ is invertible if and only if its determinant is non-zero ($|A| \neq 0$).